Chapter No. 2

Network Topologies and Networking Devices

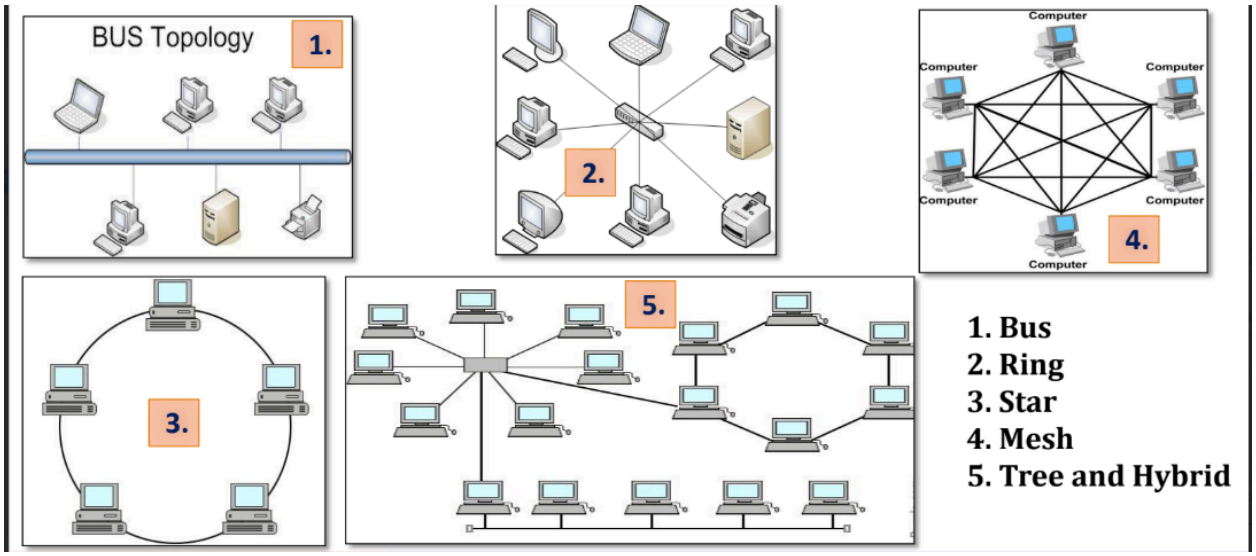
What is Network Topology?

- Geographical representation of the links.
- Topology is physical layout of computers, cables and other connected devices on a network.
- Topology defines the structure of the network of how all the components are interconnected to each other.

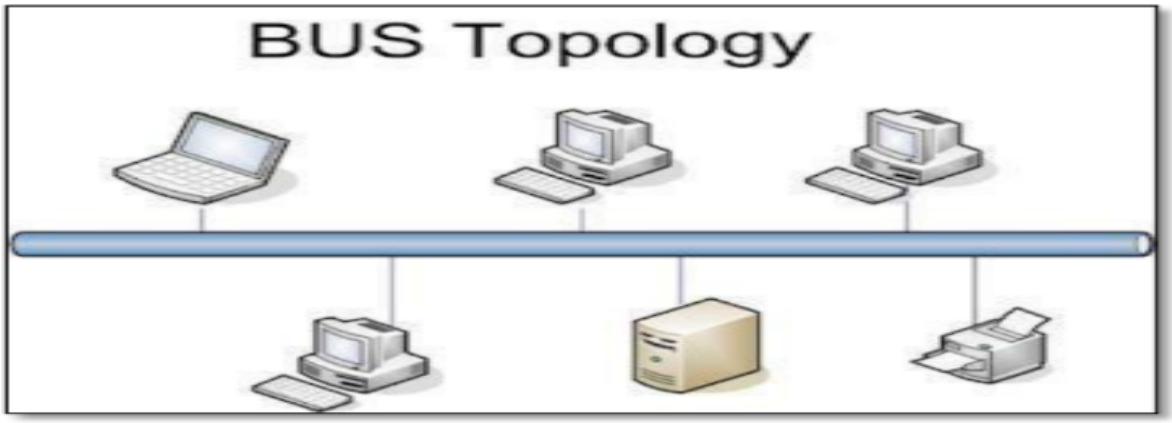
Selection Criteria

• Size (no of node) of the system.
• Cost of the components and service required.
• Management of network.
• Architecture of network.
• Cable type.
• Expandability of the network.
• The desired performance.
• Reliability.

Types of Network Topology



1. Bus Topology-



Also Called as linear topology

The bus topology is designed in such a way that all the stations are connected through a single cable known as a backbone cable.

• Each node is either connected to the backbone cable by drop cable or directly connected to the backbone cable.

• When a node wants to send a message over the network, it puts a message over the network.

• All the stations available in the network will receive the message whether it has been addressed or not. If a device want to communicate with other device on the n/w, sends a broadcast message onto the wire all other devices receive the message.

• But only the intended devices accepts and process the message.

• Bus is passive topology.

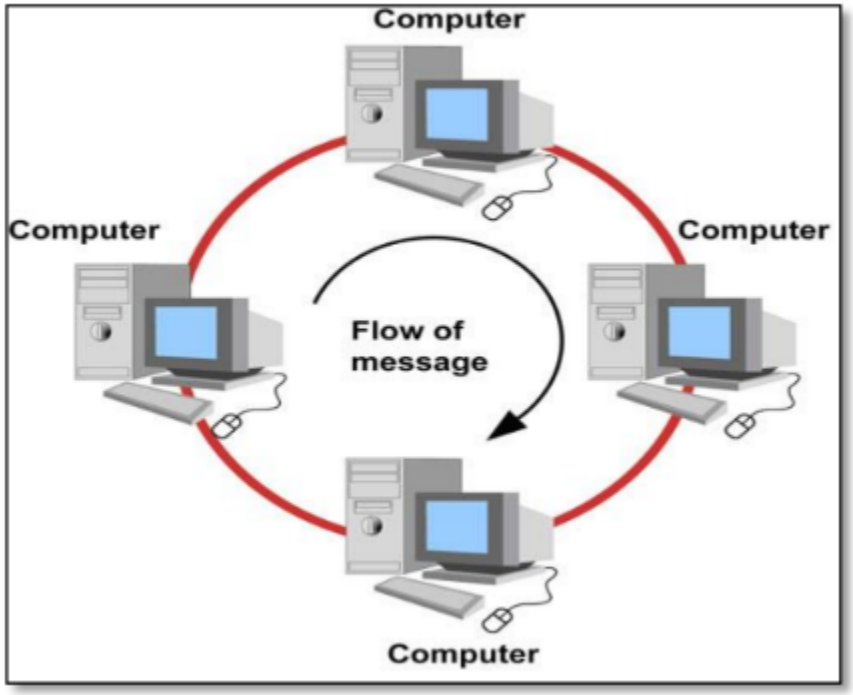
Advantages:

1. Ease of installation
2. Less cabling
3. Suitable for temporary and small networks.
4. Moderate data speeds: Coaxial or twisted pair cables are mainly used in bus-based networks that support upto 10Mbps.

Disadvantages:

1. Difficult to add new devices.
2. If any fault in backbone can stops all transmission.
3. Only one computer can transmit at a time.
4. Limited cable length and number of stations.
5. Difficult to identify the problem if the entire network shuts down
6. It is slow when more devices are added into the network
7. If a main cable is damaged then network will fail or be split into two networks

2. Ring Topology



• Each packet is sent around the ring until it reaches its final destination.
• In Ring topology each node is connected to the two nearest nodes so the entire network forms a circle.
• The signal is passed in one direction from one device to another device until it reaches the destination.
• The data flows in a single loop continuously known as an endless loop.
• It has no terminated ends, i.e., each node isconnected to other node and having no termination point.
• The data in a ring topology flow in a clockwisedirection.
• Ring is the Active Network, because eachcomputer retransmits what is receives from previous computer.

Working of Token passing:

• A token moves around the network, and it is passed from computer to computer until it reaches the destination.

• The sender modifies the token by putting the address along with the data.

• The data is passed from one device to another device until the destination address matches. Once the token received by the destination device, then it sends the acknowledgment to the sender.
• In a ring topology, a token is used as a carrier.

Advantages of Ring topology:

• Network Management: Faulty devices can be removed from the network without bringing the network down.

• Product availability: Many hardware and software tools for network operation and monitoring are available.

• Cost: Twisted pair cabling is inexpensive and easily available.
Therefore, the installation cost is very low.

• Reliable: It is a more reliable network because the communication system is not dependent on the single host computer.

Disadvantages of Ring topology:

• Difficult troubleshooting: It requires specialized test equipment to determine the cable faults. If any fault occurs in the cable, then it would disrupt the communication for all the nodes.

• Failure: The breakdown in one station leads to the failure of the overall network.

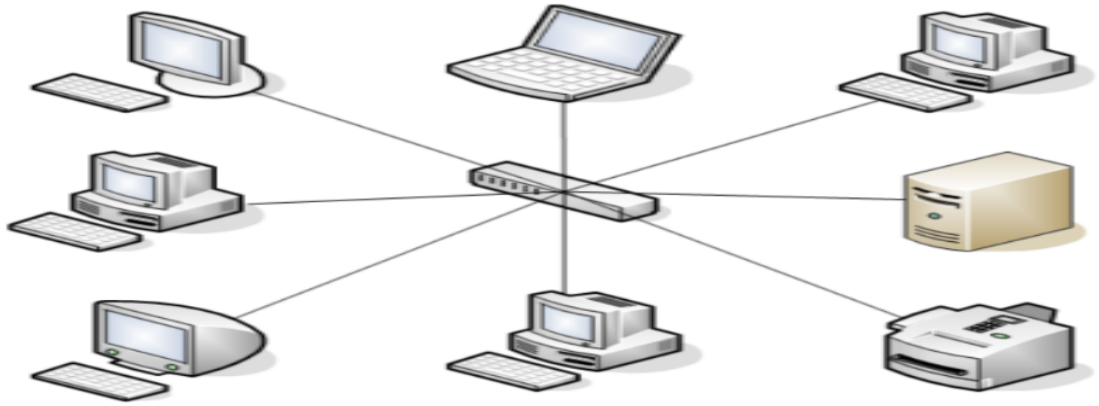
• Reconfiguration difficult: Adding new devices to the network would slow down the network.

• Delay: Communication delay is directly proportional to the number of
nodes. Adding new devices increases the communication delay.

Application

• Ring topologies are found in some office buildings or school campuses.
• Today high speed LANs made this topology less popular.

3. Star Topology



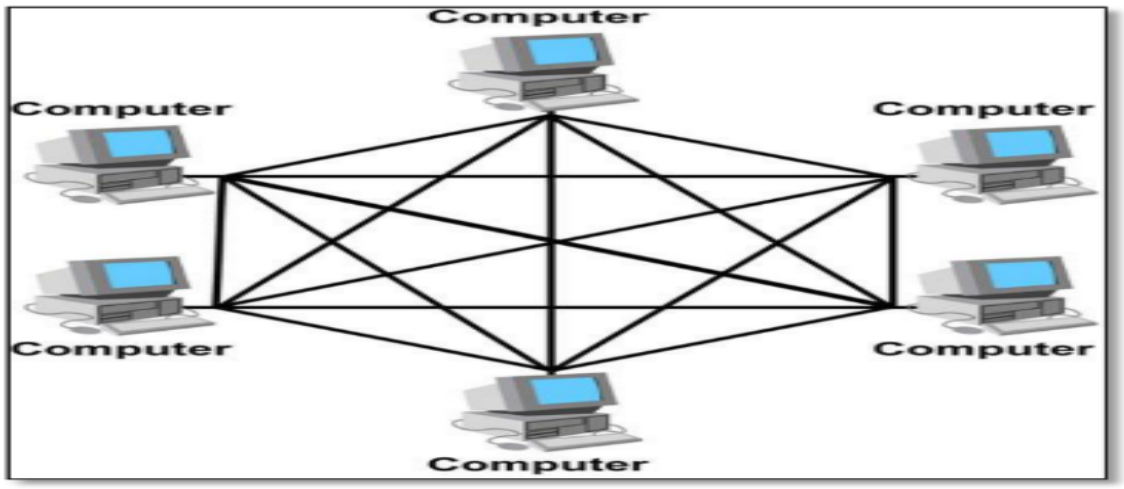
• Star topology is an arrangement of the network in which every node is connected to the central hub,switch or a central computer.
• The central computer is known as a server, and the peripheral devices attached to the server are known as clients.
• Coaxial cable or RJ-45 cables are used to connect the computers.
• Hubs or Switches are mainly used as connection devices in a physical star topology.
• Star topology is the most popular topology in network implementation.

Advantages:
• Easy to install and implement with wiring etc.
• Easy to troubleshoot and detectproblems in the network.
• If one device fails, it does not affect the other devices in the network.
• You can easily add or remove devices without affecting the rest of the network.
• Centralized management and monitoring through the central switch/hub.

Disadvantages:
• The main disadvantage of using this topology is that it has a single point of failure, i.e. when the central switch node is down, there will be a break in communication for all connected devices.
• More cabling is needed since you connect each individual devices to the central node.
• Performance of the whole network depends on the performance of the central node.

Application
• Star topology used in Local Area Networks (LANs).
• High speed LAN often use STAR.

4. Mesh Topology



• Here every device has a point to point link to every other device.
• Node-1 must be connected with n-1 nodes.
• A fully connected mesh can have n(n-1)/2 physical channels to link n devices.
• It must have n-1 input/output ports.

Advantages:

• They use dedicated links so each link can only carry its own data load. So traffic problem can be avoided.
• It is robust. If any one link get damaged it cannot affect others.
• It gives privacy and security.(Message travels along a dedicated link)
• Fault identification and fault isolation are easy.

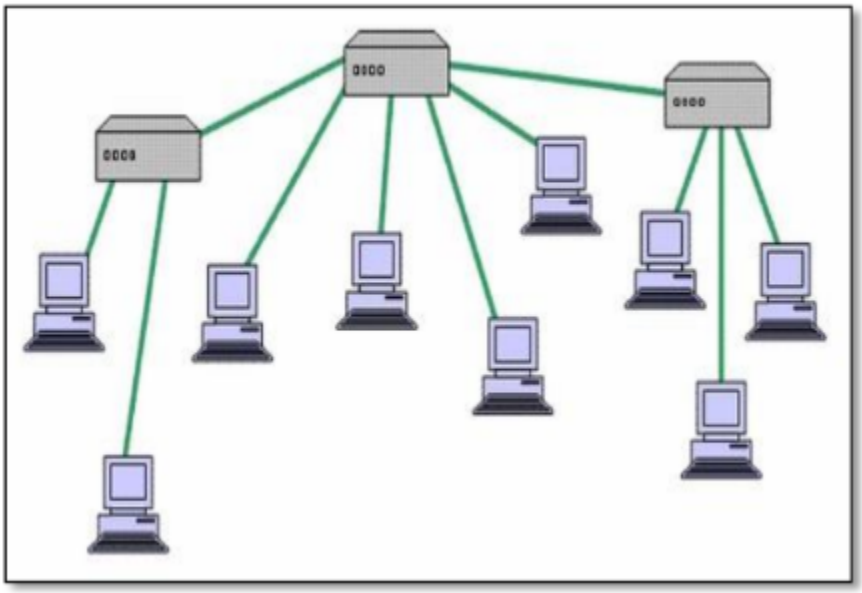
Disadvantages:

1. Cabling cost is high: The cabling and I/O ports required are more, since every device is connected to each devices through dedicated link.
2. Wiring is very complicated.
3. Hardware required to connected each device is highly expensive.
4. Troubleshooting a failed cable is tricky.

Application

• Telephone Regional office.
• WAN.(Wide Area Network).

5. Tree Topology

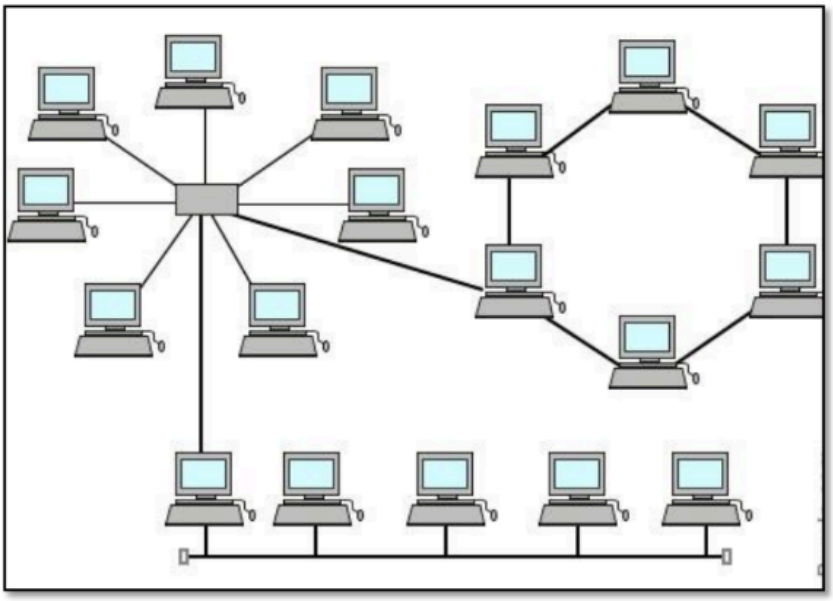


• Alternatively referred to as a star bus topology.
• This network topology is ideal for when you have a device acting as the root device on the network.
• Tree topology is one of the most common network setups that is similar to a bus topology and a star topology
• A tree topology connects multiple star networks to other star networks.

Advantages:-

• Hierarchal Model
• Scalable
• Easy Implementation
• Easy Troubleshooting

6. Hybrid Topology



• A network which contain all type of physical structure and connected under a single backbone channel.
• This network topology is basically made up of several topologies i.e.a combination of two or more topologies, which could be bus, star, ring, etc.

Advantages:
• In terms of the advantages of using this kind of topology, we find that it has a high level of fault tolerance, which makes it reliable and flexible.
• Scalable as you can add new nodes easily.
• You can design it in such a way as to use the best features of various other topologies.

Disadvantages:
• The disadvantages include things such as high cost and complexity of maintenance.
• This topology is used in real-life where scalability and high performance is important, for example Internet Service Providers.
• Costly infrastructure.

Typical Scenarios

How to choose topology and Network type

Problem Scenario 1

You are setting up a small office with 10 computers that need to share files and access a common printer.The network should be cost-effective and easy to set up while ensuring minimal disruption if one computerfails.

Ans:-

Use a Star Topology with a central switch or hub. It is easy to set up, and a single computer failure won't affect others. The network type is LAN (Local Area Network).

Problem Scenario 2

A data center needs to interconnect multiple servers to handle high-speed data transfer with redundancy to avoid failures. The network should ensure minimal latency and provide alternative paths if a connection fails.

Ans:-
Use a Mesh Topology. It provides multiple paths for data transfer and ensures high reliability. The network type is LAN or Data Center Network (DCN).

Problem Scenario 3

A remote village requires internet connectivity. The village is spread over a large area, and there are no existing wired networks nearby.

Ans:-

Use a Wireless WAN (Wide Area Network) with
technologies like satellite internet or Wi-Fi hotspots.

Problem Scenario 4

A university campus has multiple departments located in separate buildings, requiring interconnection for shared resources and scalability.

Ans:-

Use a Tree Topology combining departmental star
topologies with a central backbone. The network type is
Campus Area Network (CAN).